

A Language-Agnostic Framework for Dependency Graph Construction

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Outline

- 1 Assessing Software Quality
- 2 The "Tower of Babel" Problem
- 3 A Bit of History
- 4 A New Solution
- 5 Evaluation and Comparison
- 6 Conclusions



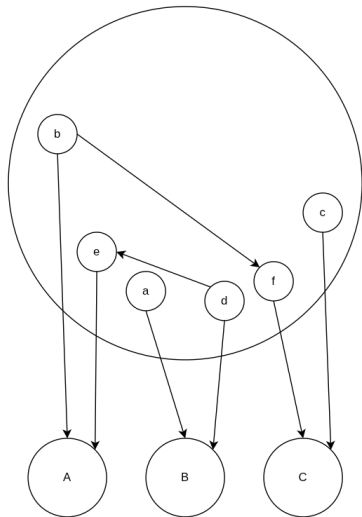
Assessing Software Quality

Definition

An **architectural smell** is a sign of poor design choices on the architectural level.

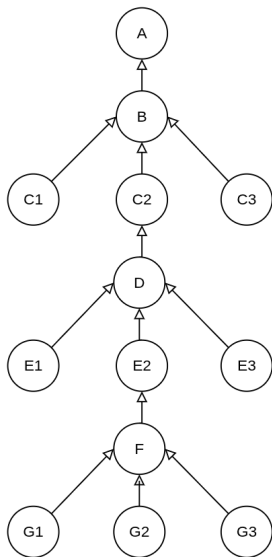
- Similar to the more known Code Smells.
- They can be detected by analyzing code shape, components and the dependency graph.
- The presence of many smells in a software project increases the estimate of **Technical Debt**, which indicates the future cost of maintenance, development and evolution.
- By Lehman's laws of software evolution we can expect it to grow over time, as bad practises effects stratify

God Component



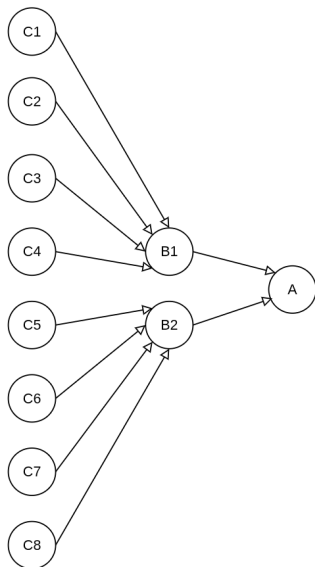
- One big component with a lot of heterogeneous sub-components inside
- Probably takes care of a lot of different concerns
- Probably has a lot of outwards dependencies and a few inwards dependencies
- Probably is a consistent percentage of the code
- Typical of Legacy Systems (especially **monolithic** architectures)

Deep Hierarchy



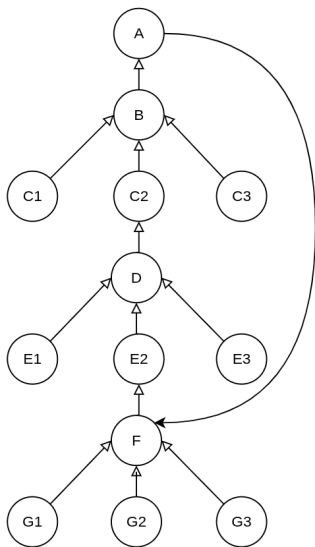
- A long inheritance chain, each level adding a bit of behaviour
- Suggests high implementation sparsity
- Suggests abuse/loss of generalization
- Typical of Object-Oriented Programming

Wide Hierarchy



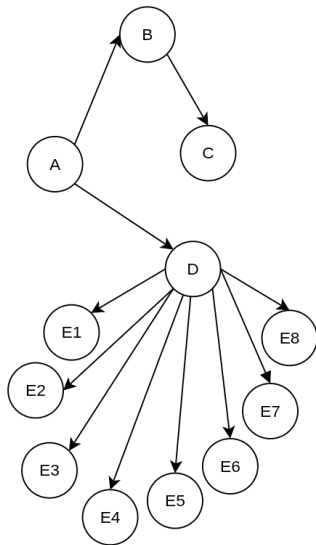
- A hierarchy chain with at each level many children
- Suggests high implementation sparsity
- Suggests abuse/loss of generalization
- Typical of Object-Oriented Programming

Cyclic Hierarchy



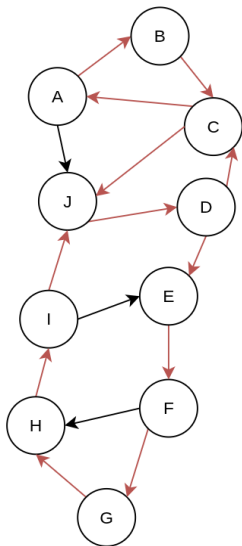
- Like the previous, but with a twist
- A parent component is actively depending on a child component
- Suggests a short-circuit of abstractions
- Typical of Object-Oriented Programming

Unstable Dependency



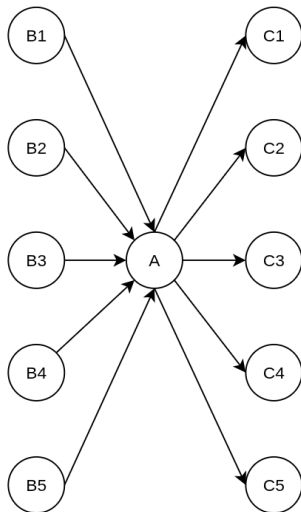
- A stable component depending on a unstable component
- (In)stability is usually defined by the ratio $\frac{C_e}{C_a + C_e}$
- Such a component is difficult to maintain stable because its dependencies have a lot of reasons to change for

Cyclic Dependency



- Components involved in a cycle are difficult to change
- Typical of legacy systems

Hub-Like Dependency



- A component with a lot of inwards and outwards dependencies
- Too stable to allow for changes
- Too unstable to depend on
- Typical of many legacy library components

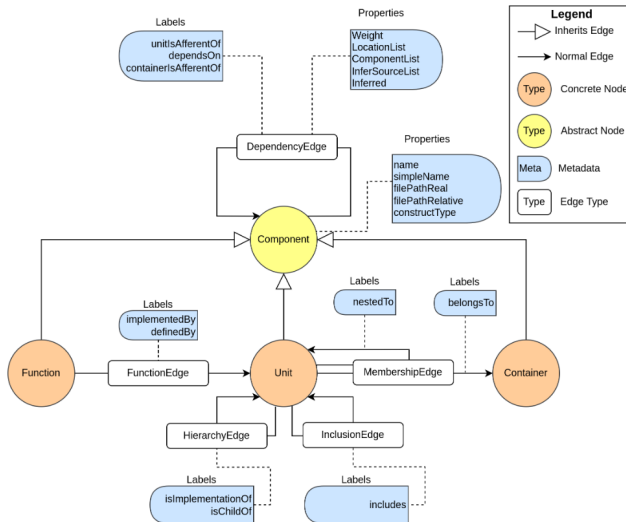
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 OVERVIEW ASSESSMENT COMPREHENSION

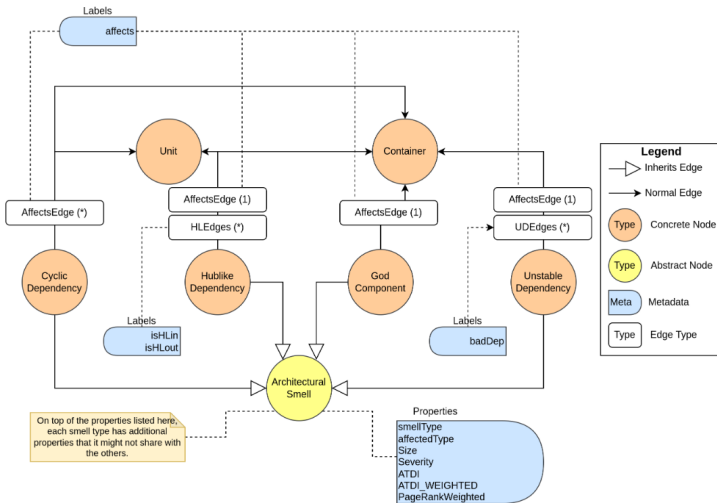
Architectural smells

ID	Smell Type	TechDebt Index	Severity	Affected Type	Affected Elements	Size
> 11606	Hublike Dependency	300	8.00	Package	org.junit.runner	21
> 11324	Cyclic Dependency	107	5.00	Package	org.junit.experimental.the...	2
> 11174	Cyclic Dependency	99	6.00	Package	org.junit.runner, org.juni...	8
> 11232	Cyclic Dependency	95	6.00	Package	org.junit.internal.runners...	8
> 11123	Cyclic Dependency	94	6.00	Package	org.junit.runner, org.juni...	6
> 11244	Cyclic Dependency	93	5.00	Package	org.junit.internal.runners...	7
> 11267	Cyclic Dependency	91	6.00	Package	org.junit.internal.runners...	8

The Dependency Graph



The Architectural Smell Graph



The "Tower of Babel" Problem

Language Segmentation

Computing language usage shares is not easy ...

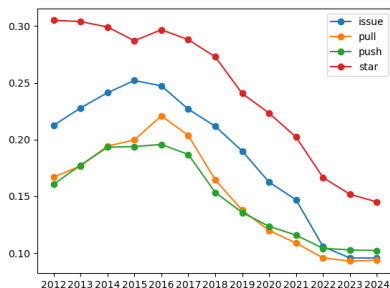


Figure: Stats of JavaScript

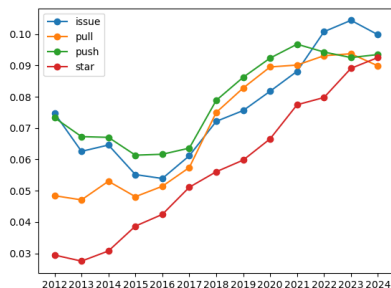
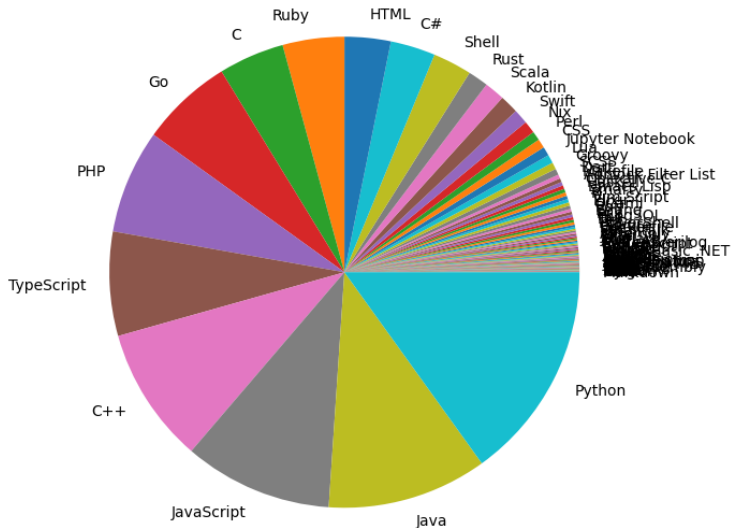


Figure: Stats of C++

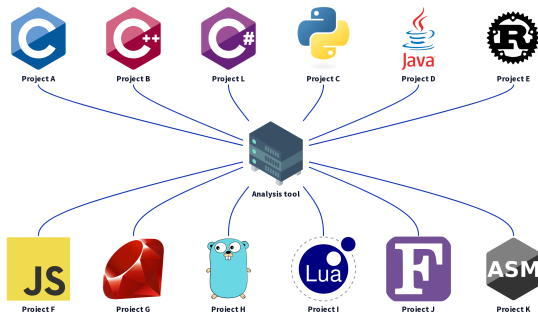
Language Segmentation

... but we can do some estimates



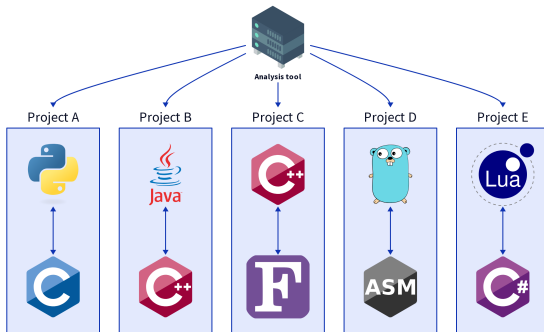
A matter of scale

- A software analysis tool need to work with many programming languages.
- Each deployed instance need every language interface to be useful
- Each language interface is a reason to change and increases maintenance costs



A matter of perspective

- A project can use more than one language, which is usually not supported by tools
- Each language interface usually need a different implementation for the same logic
- Sometimes it is easier to develop a distinct version of each language



A Bit of History

Language Frontends

- Common approach: use language frontends (e.g., Clang, Eclipse JDT/CDT) to parse code and extract data.
- APIs are heterogeneous and lack standardization.
- The *Language Server Protocol (LSP)* aims to unify such interfaces, but still has limits in query expressiveness and flexibility.
- Frontends are designed for IDEs/compiler, requiring extra adaptation for analysis tools.
- Typically heavy-weight — include many elaborations not needed by analysis tools.

Manual/Automatic Transpilation

- Some works avoid using multiple language frontends by translating source code into a single target language or DSL.
- Translation can be done manually or via transpilers.
- Works well for languages designed for transpilation (e.g., TypeScript, Haxe).
- For other languages, translation may lose semantic details (e.g., classes, namespaces, packages).
- To preserve semantics, the target language must be as expressive as the source.
- This approach remains inefficient:
 - Automatic translation between languages is still an open research problem.
 - Manual translation is time-consuming and error-prone.
 - Translated code must still be parsed and analyzed afterward.

AST/CST Translation

- Another approach: translate a **language-dependent AST** into a **language-independent AST**.
- Enables analysis tools to apply algorithms on a common, unified representation.
- Can also be applied to **Concrete Syntax Trees (CSTs)**, which capture full program syntax.
- Typically relies on language frontends (manual or generated) to perform near 1:1 translation into an *extended AST (eAST)*.
- Sometimes produces an intermediate serialized form (e.g., XML).

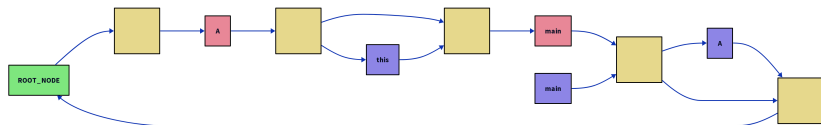
- A **meta model abstraction** offers a language-agnostic alternative:
 - Represents high-level program entities — classes, structs, functions — instead of raw syntax trees.
 - Simplifies cross-language analysis by focusing on shared structural concepts.
 - Reduces dependency on language frontends and heavy parsing.
- Enables more flexible and extensible analysis frameworks built around **semantic equivalence** rather than syntax.
- Often requires custom frontends to build the meta-model from source code.

- Beyond traditional frontends, new data structures have been designed for specific analysis tasks (e.g., code navigation, reference resolution).
- Evaluated the use of **Stack Graphs** — composable graphs representing identifiers and their relationships — for language-independent reference resolution.
- Showed promising accuracy but faced:
 - Practical and scalability limitations.
 - Partial language-independence: graph construction still required language-specific CST queries.
- Despite limitations, the study highlighted:
 - The potential of **language-independent analysis**.
 - Its value as a guiding principle for future software analysis strategies.

ADS / Stack Graphs

- Composable graph
- Represents identifiers and scopes of source code
- Enables language agnostic reference resolution

```
class A {  
    public static void main() {  
    }  
}
```



ADS / Tree Sitter Stack Graphs

- A "TSSG" Grammar is composed of pairs
- Each pair matches a section of the CST with some procedural code
- Definition of nodes, edges and labels

```
(field_access object: (_)@object field: (_)@field)@this {  
  node @this.scope  
  node @this.pop_start  
  node @this.pop_end  
  edge @field.push_end -> @object.push_start  
  let @this.push_start = @field.push_start  
  let @this.push_end = @object.push_end  
  edge @this.push_end -> @this.scope  
  edge @this.pop_start -> @this.push_start  
  attr (@field.push_start)  
    is_reference, refkind = "accessField"  
}
```

A New Solution

Very different ...

```
import numpy

def dct(X: numpy.ndarray) -> numpy.ndarray:
    if len(X.shape) == 1:
        M = X.shape[0]
        return dct(X.reshape((1, M)).reshape((M,)))

    N, M = X.shape
    Y: numpy.ndarray = numpy.zeros((N, M), dtype=numpy.float64)

    pi_over_M = numpy.pi / M
    sqrt_of_2_over_M = numpy.sqrt(2.0 / M)
    one_over_sqrt_of_two = numpy.sqrt(0.5)
    j_minus_half_from_1_to_M = (numpy.array(range(M)) + 0.5)

    for i in range(0, N):
        for k in range(0, M):
            k_pi_over_M = k * pi_over_M
            Y[i, k] = sum(
                numpy.cos(
                    k_pi_over_M *
                    j_minus_half_from_1_to_M *
                    X[i]) * sqrt_of_2_over_M
            )
            Y[i, 0] = Y[i, 0] * one_over_sqrt_of_two
        return Y

def dct2(X: numpy.ndarray) -> numpy.ndarray:
    return numpy.transpose(dct(numpy.transpose(dct(X))))
```

```
#include <dcct/slow_actuator.hh>

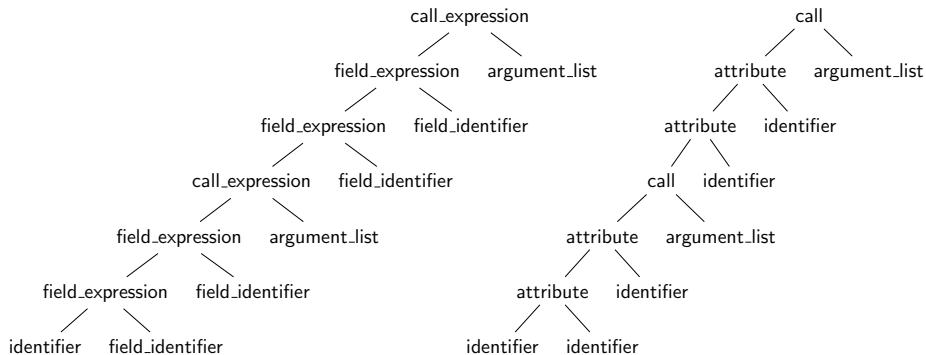
Eigen::MatrixXd dcct::SlowActuator::dct(Eigen::MatrixXd& X) {
    uint32_t N = X.rows(), M = X.cols();
    Eigen::MatrixXd Y(N, M);

    double_t pi_over_M = std::numbers::pi / (double_t) M;
    double_t sqrt_of_two_over_M = std::sqrt(2.0 / (double_t) M);
    double_t one_over_sqrt_of_two = std::sqrt(0.5);

    for (uint32_t i = 0; i < N; ++i) {
        for (uint32_t k = 0; k < M; ++k) {
            double_t sum = 0;
            double_t _k_pi_over_M = pi_over_M * k;
            for (uint32_t j = 0; j < M; ++j) {
                sum += std::cos(_k_pi_over_M * (j + 0.5)) * X.coeff(i, j);
            }
            Y.coeffRef(i, k) = sum * sqrt_of_two_over_M;
        }
        Y.coeffRef(i, 0) *= one_over_sqrt_of_two;
    }
    return Y;
}

Eigen::MatrixXd dcct::SlowActuator::dct2(Eigen::MatrixXd& X) {
    Eigen::MatrixXd Y = dct(X).transpose();
    X = dct(Y).transpose();
    return X;
}
```

... Or very similar?



Tree Sitter



- General enough to parse any programming language
 - Fast enough to parse on every keystroke in a text editor
 - Robust enough to provide useful results even in the presence of syntax errors
 - Dependency-free so that the runtime library (which is written in pure C11) can be embedded in any application
- Tree-sitter is a parser generator tool and an incremental parsing library.
 - It can build a concrete syntax tree for a source file and efficiently update the syntax tree as the source file is edited.

Why Tree Sitter?

Tree Sitter is not only easy to operate, but it is also easy to implement parsers for new languages.

```
typedef [0, 0] - [0, 32]
  name: identifier [0, 8] - [0, 12]
  type: integer_type [0, 15] - [0, 31]
    size: integer [0, 23] - [0, 24]
    signed: boolean [0, 26] - [0, 30]
function [1, 0] - [1, 32]
  name: identifier [1, 3] - [1, 9]
  parameters: parameter_list [1, 9] - [1, 31]
    parameter [1, 10] - [1, 20]
      name: identifier [1, 10] - [1, 13]
      type: pointer_type [1, 15] - [1, 20]
        type: identifier [1, 16] - [1, 20]
    parameter [1, 22] - [1, 30]
      name: identifier [1, 22] - [1, 23]
      type: pointer_type [1, 25] - [1, 30]
        type: identifier [1, 26] - [1, 30]
```

Figure: Example of Lart CST

```
typedef char = integer<8, true>;
fn printf(fmt: &char, s: &char);
```

Figure: Example of Lart code

```
function: $ => seq(
  'fn',
  field('name', $.identifier),
  field('parameters', $.parameter_list),
  optional(seq('>', field('type', $_.type))),
  choice(field('body', $.block), ';')
),

include: $ => seq(
  'include',
  choice(
    seq('<', field('globalpath', $.path_literal), '>'),
    seq('"', field('localpath', $.path_literal), '"')
  ), ';'
),
```

Figure: Slice of Lart grammar as TS config

The Approach

Development

- Gain initial knowledge of a language's **grammar**
- Gain initial knowledge of a language's **semantics**
- Prepare a **language benchmark** to verify that the implementation is correct
- Each programming language shall be described as a **feature set**

Operation

- Extract CSTs with **Tree Sitter** for each file
- Traverse the CSTs to capture the components definitions and the references
- Construct a *Language-Independent* **Component Tree**
- Accumulate the **Dependency Queries**
- Resolve the Dependency Queries with the Component Tree into the **Dependency Graph**
- Export the Dependency Graph to JSON

How it works / CST Traversal / Analysis

Java

```
(method_declaration
  type: (void_type)
  name: (identifier)
  parameters: (formal_parameters)
  body: (block))
```

Rust

```
(function_item
  name: (identifier)
  parameters: (parameters)
  return_type: (unit_type)
  body: (block))
```

C++

```
(function_definition
  type: (primitive_type)
  declarator: (function_declarator
    declarator: (identifier)
    parameters: (parameter_list))
  body: (compound_statement))
```

Python

```
(function_definition
  name: (identifier)
  parameters: (parameters)
  return_type: (type (none))
  body: (block (pass_statement)))
```

How it works / CST Traversal / Translation

Java

```
if tsnode.type == "method_declaration" {  
  let function = Function {  
    name: process_name(tsnode.get_field("name")),  
    type: process_type(tsnode.get_field("type")),  
  }  
  process_parameters(tsnode.get_field("parameters"));  
  process_body(tsnode.get_field("body"));  
}
```

Rust

```
if tsnode.type == "function_item" {  
  let function = Function {  
    name: process_name(tsnode.get_field("name")),  
    type: process_type(tsnode.get_field("return_type")),  
  }  
  process_parameters(tsnode.get_field("parameters"));  
  process_body(tsnode.get_field("body"));  
}
```

C++

```
if tsnode.type == "function_definition" {  
  let declarator = tsnode.get_field("declarator");  
  let function = Function {  
    name: process_name(declarator.get_field("declarator")),  
    type: process_type(tsnode.get_field("type")),  
  }  
  process_parameters(declarator.get_field("parameters"));  
  process_body(tsnode.get_field("body"));  
}
```

Python

```
if tsnode.type == "function_definition" {  
  let function = Function {  
    name: process_name(tsnode.get_field("name")),  
    type: process_type(tsnode.get_field("return_type")),  
  }  
  process_parameters(tsnode.get_field("parameters"));  
  process_body(tsnode.get_field("body"));  
}
```

How it works / CST Traversal / Generalization

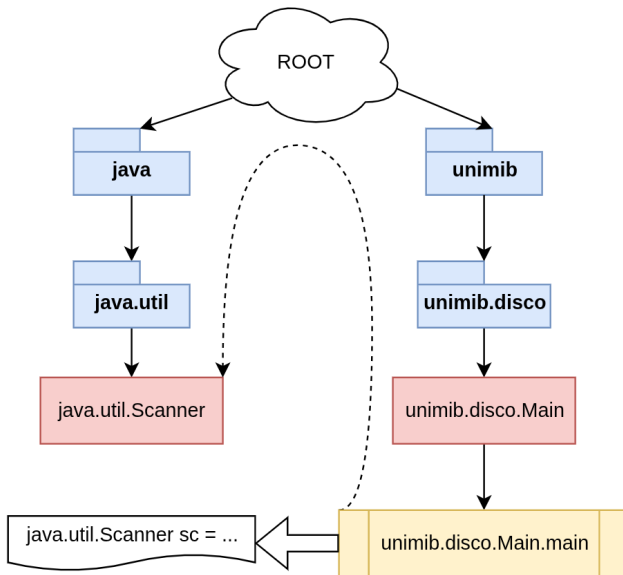
Java / Rust / Python

```
if tsnode.type == Lang.tags.defun {  
  let function = Function {  
    name: process_name(tsnode.get_field(Lang.tags.defun.direct.name)),  
    type: process_type(tsnode.get_field(Lang.tags.defun.direct.type)),  
  }  
  process_parameters(tsnode.get_field(Lang.tags.defun.direct.parameters));  
  process_body(tsnode.get_field(Lang.tags.defun.direct.body));  
}
```

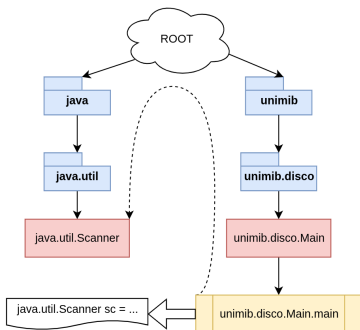
C / C++

```
if tsnode.type == Lang.tags.defun {  
  let declarator = tsnode.get_field(Lang.tags.defun.declarator.declarator);  
  let function = Function {  
    name: process_name(declarator.get_field(Lang.tags.defun.declarator.name)),  
    type: process_type(tsnode.get_field(Lang.tags.defun.declarator.type)),  
  }  
  process_parameters(declarator.get_field(Lang.tags.defun.declarator.parameters));  
  process_body(tsnode.get_field(Lang.tags.defun.declarator.body));  
}
```

How it works / Dependency Resolution / Identifier Chains



How it works / Dependency Resolution / Identifier Chains

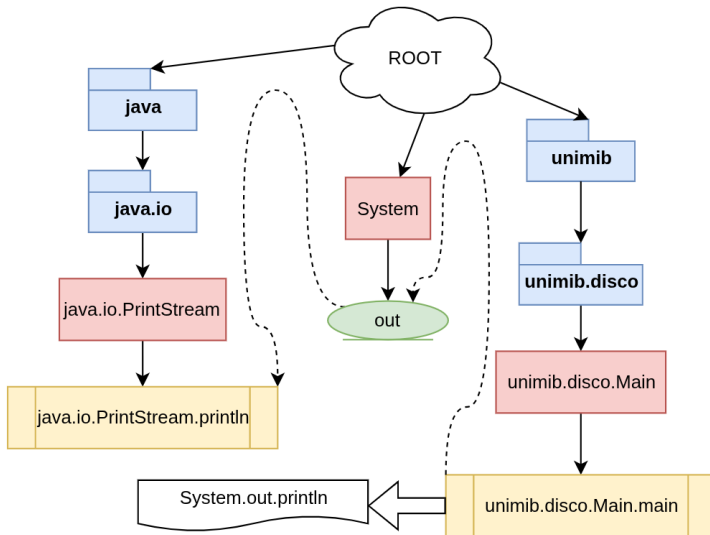


```
def resolve(scope: Node, reference: list[str]):
    assert len(reference) > 0
    scope = find_parent(reference.pop())
    while len(reference) > 0 and scope is not None:
        scope = find_child(scope, reference.pop())
    return scope
```

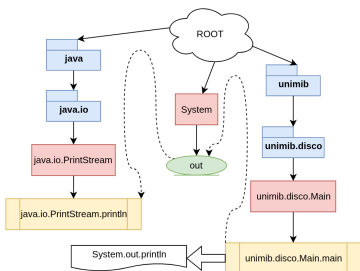
```
def find_parent(scope: Node, identifier: str):
    while scope is not None:
        for child in children(scope):
            if name(child) == identifier:
                return child
        scope = parent(scope)
    return None
```

```
def find_child(scope: Node, identifier: str):
    for child in children(scope):
        if name(child) == identifier:
            return child
    return None
```

How it works / Dependency Resolution / Type Resolution



How it works / Dependency Resolution / Type Resolution



```
def resolve(scope: Node, reference: list[str]):
    assert len(reference) > 0
    scope = find_parent(reference.pop())
    while len(reference) > 0 and scope is not None:
        if kind(scope) == FIELD:
            scope = type(scope)
            scope = find_child(scope, reference.pop())
    return scope
```


How it cheats / Ghost Nodes

Problem

- Code definitions are caught through CST traversal
- References can be solved iff both source and target are on the graph
- TPL's components cannot be detected because their code isn't read
- References TPL's components cannot be detected because they aren't on the graph

Example

```
import fi.foyt.foursquare.api.FoursquareApi;
```

Solution

- Every time we should find a component but nothing is there
- Ex: a method 'foo' inside class 'FoursquareApi' which isn't there
- Create a 'ghost' node with volatile a kind so that we can decided what to do

How it cheats / Operator Overloading

Problem

- Some programming languages allow operator overloading
- Syntactically, these are treated as binary/monary operators
- Overloaded operators carry complex logic

Example

```
void main() {  
    Vector3 A = Vector3 {0,0,1}, B = Vector3 {1,0,0};  
    Vector3 C = A + B; // A.operator+(B)  
}
```

Solution

- Usually, programming languages define overloading as methods
- We can intercept those method definitions knowing their encoded name
- We can guess their name since they usually have a naming schema
- This data can be added to the language feature set

How it cheats / Deref Trait

Problem

- Rust and other languages allow definitions of Delegate Operands
- These delegates are not defined as parent classes or target of implementations
- Languages don't offer clear indicators at component definition level

Example

```
impl Deref for StructC { type Target = StructA;  
    fn deref(&self) -> &Self::Target { &self.0 }}  
fn main() {  
    StructC::new().method();  
}
```

Solution

- Usually these are implemented as methods or attributes
- Knowing how they are named, we can intercept them

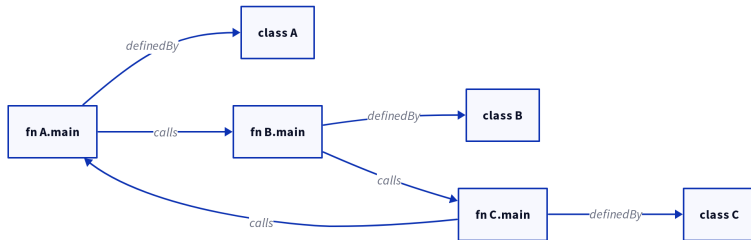
An Example / The Code

```
class A {  
    public static void main() {  
        B.main();  
    }  
}
```

```
class B {  
    public static void main() {  
        C.main();  
    }  
}
```

```
class C {  
    public static void main() {  
        A.main();  
    }  
}
```

An Example / The Dependency Graph



Evaluation and Comparison

The Testing Framework

Tests are described with YAML manifests including constraints on nodes and edges that must be detected in the dependency graph written by the tool

```
filepaths:
  - main.java
nodes:
  - name: ClassName
    kind: class
  - name: ClassName.MethodName
    kind: function
edges:
  - source: ClassName.MethodName
    sink: ClassName
    kind: definedBy
```

- node types: class, function, enum ...
- edge types: definedBy, includes, usesType, calls ...

The Dependency Detection Benchmark

Dependency type	Example code
Import	
Class import	<code>import ModuleB.ModuleB2.Class3;</code>
Declaration	
Instance variable	<code>private Class3 class3;</code>
Class variable	<code>private static Class3 class3;</code>
Local variable	<code>public void method() Class3 class3;</code>
Parameter	<code>public void method(Class3 class3)</code>
Return type	<code>public Class3 method()</code>
Exception	<code>public void method() throws Class4throw new Class4 ("...");</code>
Type cast	<code>Object o = (Class3) new Object();</code>
Call	
Instance method	<code>variable = class3.method();</code>
Instance method-inherited	<code>variable = class3.methodSuper();</code>
Class method	<code>variable = class3.classMethod();</code>
Constructor	<code>new Class3();</code>
Inner class	<code>method variable = class3.InnerClass.method();</code>
Interface method	<code>interface I.interfaceMethod();</code>
Library class method	<code>libraryClass1.libraryMethod();</code>
Access	
Instance variable	<code>variable = class3.variable;</code>
Instance variable-inherited	<code>variable = class3.variableSuper;</code>
Class variable	<code>variable = Class3.classVariable;</code>
Constant variable	<code>variable = class3.constantVariable;</code>
Enumeration	<code>System.out.println(Enumeration.VAL1);</code>
Object reference	<code>method(class3);</code>
Inheritance	
Extends class	<code>public class Class1 extends Class3</code>
Extends abstract	<code>class Idem but in this case Class3 should be abstract.</code>
Implements interface	<code>public class Class1 implements Interface1</code>
Annotation	
Class annotation	<code>@Class3</code>

The Dependency Detection Benchmark

Dependency type	Example code
Call	
Instance method	<code>variable = class2.class3.method();</code>
Instance method-inherited	<code>variable = class2.methodSuper();</code>
Class method	<code>variable = class2.class3.classMethod();</code>
Access	
Instance variable	<code>variable = class2.class3.variable;</code>
Instance variable-inherited	<code>variable = class2.variableSuper();</code>
Class variable	<code>variable = class2.class3.classVariable;</code>
Object reference-Reference var	<code>variable = class2.method(class2.class3.class4);</code>
Object reference-Return value	<code>Object o = (Object) class2.getClass4();</code>
Inheritance	
Extends-implements variations	<code>public class Class1 extends Class2</code> <code>public class Class2 extends SuperClass</code>

1

¹The accuracy of dependency analysis in static architecture compliance checking, Pruijt et Al, Softw. Pract. Exp. 2017

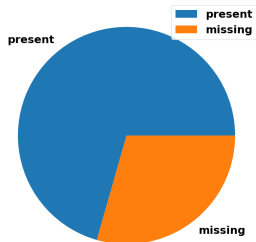


Figure: Adherence of Arcan

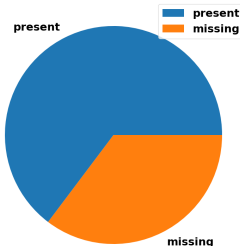


Figure: Adherence of the Stack-Graph based solution

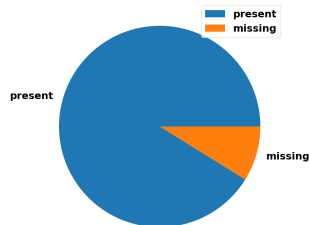
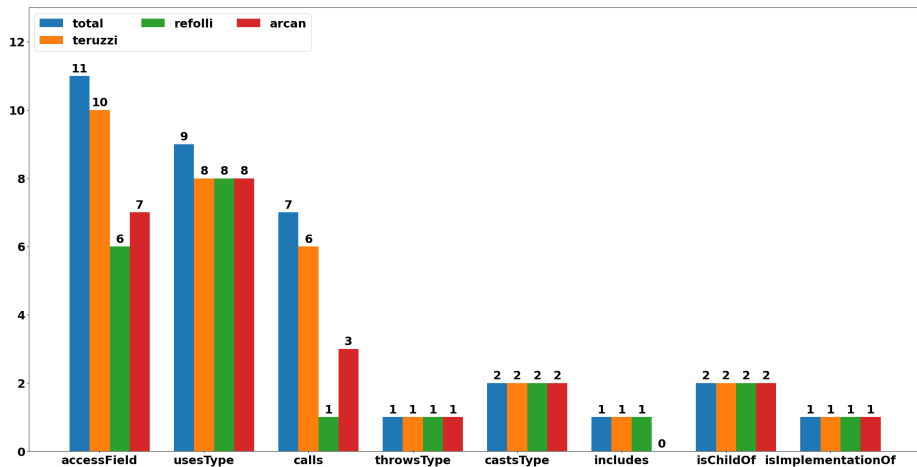
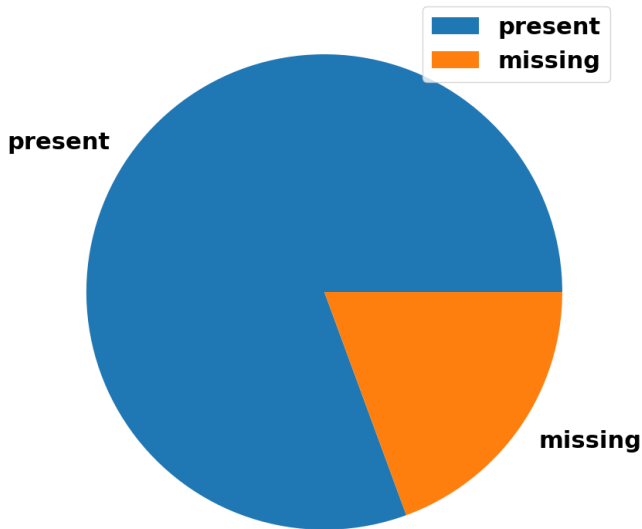
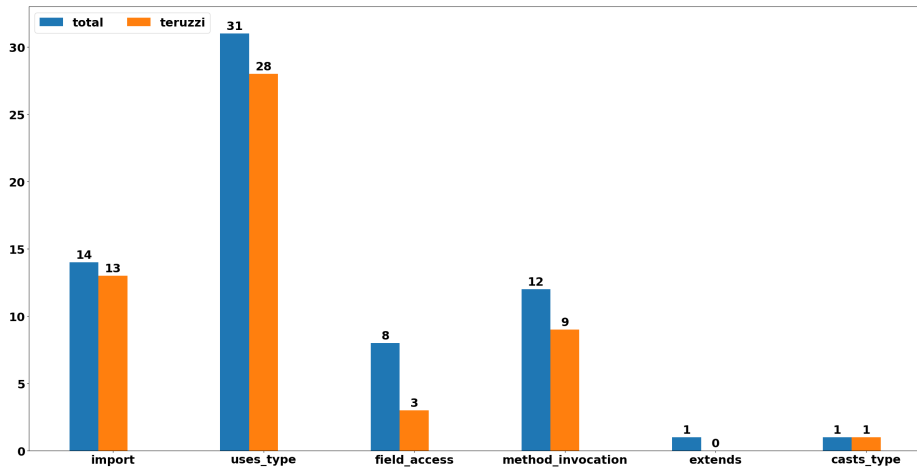


Figure: Adherence of the new proposed solution





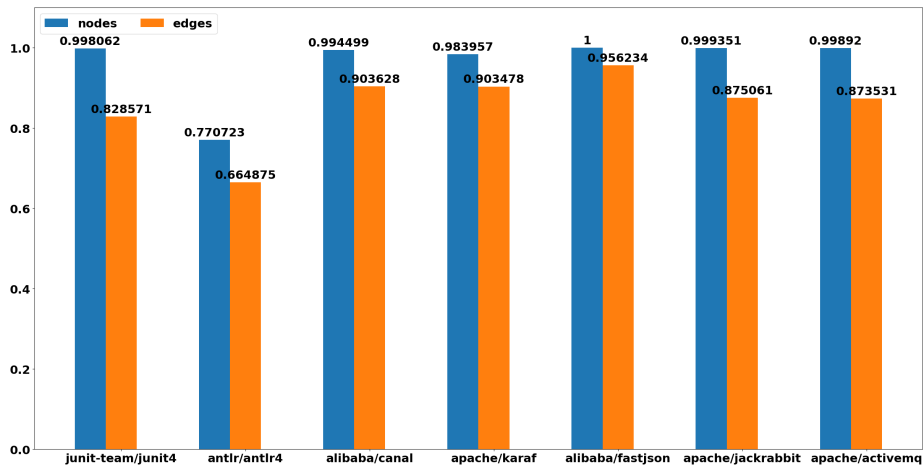


Comparison with the previous iteration

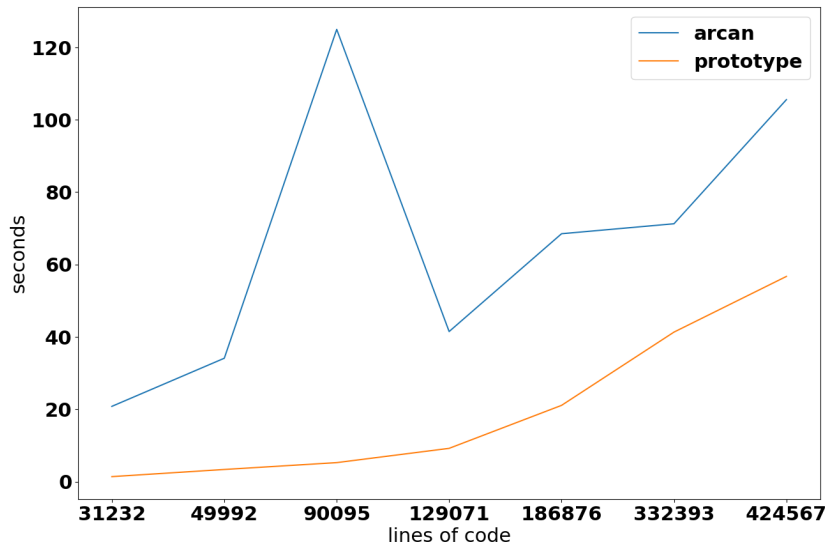
- The Stack-Graph based solution had a decent effectiveness when dealing with small scale projects, but efficiency is another story.
- In some cases it didn't even complete the scan within a reasonable time.
- For these reasons, the comparison is drawn mainly with Arcan in the next slides.

Project	Tool	Min	Max	Mean execution time
JUnit4	Stack-Graph	65,82	69,07	65,88
JUnit4	Arcan	13,850	17,079	14,611
JUnit5	Stack-Graph	132,87	134,75	134,44
JUnit5	Arcan	42,542	47,613	44,186
ANTLR	Stack-Graph	171,65	175,49	172,64
ANTLR	Arcan	19,222	20,140	19,691
Fastjson	Stack-Graph	N/A	N/A	N/A
Fastjson	Arcan	66,932	71,094	69,071

Comparison with Arcan / Similarity



Comparison with Arcan / Efficiency



Conclusions

Open Problems

Future Works

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